

PAEDIATRIC EMERGENCY DEPARTMENT, HOSPITAL TUNKU AZIZAH
ASTHMA ORDER SET

Date :	Time:	Age:	Weight:	No. of encounter:
Name :				
Address :			Contact No:	
D.O.B :	Sex: M / F	IC/Passport:	MRN:	
Allergy:		Co-morbid:		

☐ Inform MO for patients < 2 years old

☐ Hx of intubation / ICU / HDW admission

*** Refer page 3 for PAS parameters and Drug dilutions**
page 4 for Asthma Algorithm

1. Pretreatment. Date: _____ Time: _____ PAS: _____ PEFR: _____	Time noted : MO/SN/PPP sign
2. Initial Orders	
☐ PAS >12 : Apply continuous cardiorespiratory monitoring ☐ SpO2 <95% : Consider supplemental oxygen ☐ PAS every 20 mins or earlier ☐ PAS >8 : Consult Medical Officer/Emergency Physician	BP: _____ mmHg Heart rate: _____ x/min RR: _____ x/min O2 sat: _____ % Lungs: _____ PEWS+ Score: _____
3. Initial Treatment	
A. Bronchodilator (COMPULSARY)	
☐ _____ Mild (5 - 7)	MDI Salbutamol + Spacer ≤6 yo: 4-6 puffs >6 yo: 8-10 puffs.
☐ _____ Moderate (8 – 11)	MDI Salbutamol + Spacer (Every 20 mins interval, max 3 doses) (Every 20 mins interval, max 3 doses) ☐ ≤6 yo: 4-6 puffs ☐ >6 yo: 8-10 puffs
☐ _____ Severe/life threatening (12 – 15)	MDI Salbutamol + Spacer ☐ ≤6 yo: 4-6 puffs ☐ >6 yo: 8-10 puffs MDI Ipratropium Bromide + Spacer ☐ ≤6 yo: 4 puffs ☐ >6yo: 8 puffs } Every 20 mins interval, max 3 doses OR *If life-threatening or hypoxic Nebulised Salbutamol : Atrovent : NS (Back to back X 3 doses within first hour) ☐ ≤5 yo: Salbutamol 2.5mg (0.5ml): Atrovent 250mcg (1ml): NS 2.5ml ☐ >5 yo: Salbutamol 5mg (1ml): Atrovent 500mcg (2ml): NS 1ml
B. Corticosteroid (COMPULSARY)	
All Patients	☐ Oral Prednisolone (1mg/kg/day (max 30-40mg) Dose: _____mg stat given at _____ for 3 - 5 days OR ☐ IV Hydrocortisone stat [4mg/kg (max 100mg)] (if there is contraindication for oral prednisolone or altered mental status) -> Dose: _____mg given at _____
4. Additional Treatment	
☐ Oxygen: _____ ☐ Reassess in _____ min. ☐ Uptriage to _____ zone * for Severe/Life Threatening ☐ Continuous cardiorespiratory monitoring ☐ IV Salbutamol continuous infusion 1-5mcg/kg/min, (Loading 5-15mcg/kg over 10 mins) -> IV Salbutamol loading _____ mcg/kg over 10 mins, then _____ mcg/kg/min. ☐ IV Magnesium Sulphate 50% bolus 0.1ml/kg (50mg/kg, max 2500mg/dose) -> IV Magnesium Sulphate 50% _____ mg over 20 mins. Oxygen supplementation: ☐ +/- Non-invasive / Mechanical Ventilation. Setting: _____ ☐ Consider HDU/ICU admission	

5. Reassessment Pathway		
A. First Reassessment at 20 mins. Date: _____ Time: _____ PAS: _____ PEFr: _____		
☐ Mild (5 -7)	☐ Continue observation. Reassess 60 minutes after last dose of Salbutamol for disposition.	
☐ Moderate (8 – 11)	MDI Salbutamol + Spacer: 3 DOSES (To complete with 20 minutes interval within first hour) ☐ ≤6 yo: 4-6 puffs ☐ >6 yo: 8-10 puffs	
☐ Severe/life threatening (12 – 15)	MDI Salbutamol + Spacer ☐ ≤6 yo: 4-6 puffs ☐ >6 yo: 8-10 puffs Every 20 mins interval, max 3 doses If not given on arrival, MDI Ipratropium Bromide + Spacer ☐ ≤6 yo: 4 puffs ☐ >6 yo: 8 puffs OR *If life-threatening or hypoxic Nebulised Salbutamol : Atrovent : NS (Back to back X 3 doses within first hour) ☐ ≤5 yo: Salbutamol 2.5mg (0.5ml): Atrovent 250mcg (1ml): NS 2.5ml ☐ >5 yo: Salbutamol 5mg (1ml): Atrovent 500mcg (2ml): NS 1ml ☐ Call senior for help <u>If not given yet</u> ☐ IV Magnesium sulphate 50% bolus 0.1ml/kg (50mg/kg/dose, max 2500mg/dose) IV Magnesium Sulphate 50% ____ mg over 20 mins ☐ IV Salbutamol continuous infusion 1-5mcg/kg/min, (Loading 5-15mcg/kg over 10 mins) IV Salbutamol loading ____ mcg over 1o mins, then ____ mcg/kg/min.	

B. Second Reassessment within 1 hour. Date: _____ Time: _____ PAS: _____ PEFR: _____

☐ Moderate (8 – 11)	MDI Salbutamol + Spacer (to complete 3 doses within first hour, then every 4 hourly) ☐ Oxygen: _____ ☐ Reassess in _____ min. ☐ Monitor for worsening respiratory distress ☐ Uptriage to zone: _____ ☐ ≤6 yo: 4-6 puffs ☐ >6 yo: 8-10 puffs. } Every 20 mins interval, max 3 doses	
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☐ Severe/life threatening (12 – 15)	MDI Salbutamol + Spacer ☐ ≤6 yo: 4-6 puffs ☐ >6 yo: 8-10 puffs. <u>If not given on arrival,</u> MDI Ipratropium Bromide + Spacer ☐ ≤6 yo: 4 puffs ☐ >6 yo: 8 puffs	Every 20 mins interval, max 3 doses OR	*If life-threatening or hypoxic Nebulised Salbutamol : Atrovent : NS (Back to back X 3 doses within first hour) ☐ ≤5 yo: Salbutamol 2.5mg (0.5ml): Atrovent 250mcg (1ml): NS 2.5ml ☐ >5 yo: Salbutamol 5mg (1ml): Atrovent 500mcg (2ml): NS 1ml
	☐ +/- Non-invasive / Mechanical Ventilation. Setting: _____ ☐ Oxygen 10L/min by HFM ☐ MDI/Neb Salbutamol: _____ (dose/frequency) ☐ Continuous cardiorespi. monitoring ☐ Monitor for worsening respiratory distress ☐ Call senior for help ☐ Treat in Critical Zone <u>If not given yet</u> ☐ IV Magnesium sulphate 50% bolus 0.1ml/kg (50mg/kg/dose, max 2500mg/dose) IV Magnesium Sulphate 50% ____ mg over 20 mins ☐ IV Salbutamol continuous infusion 1-5mcg/kg/min, (Loading 5-15mcg/kg over 10 mins) IV Salbutamol loading ____ mcg over 10 mins, then ____ mcg/kg/min. ☐ Consider HDU/ICU admission		

2

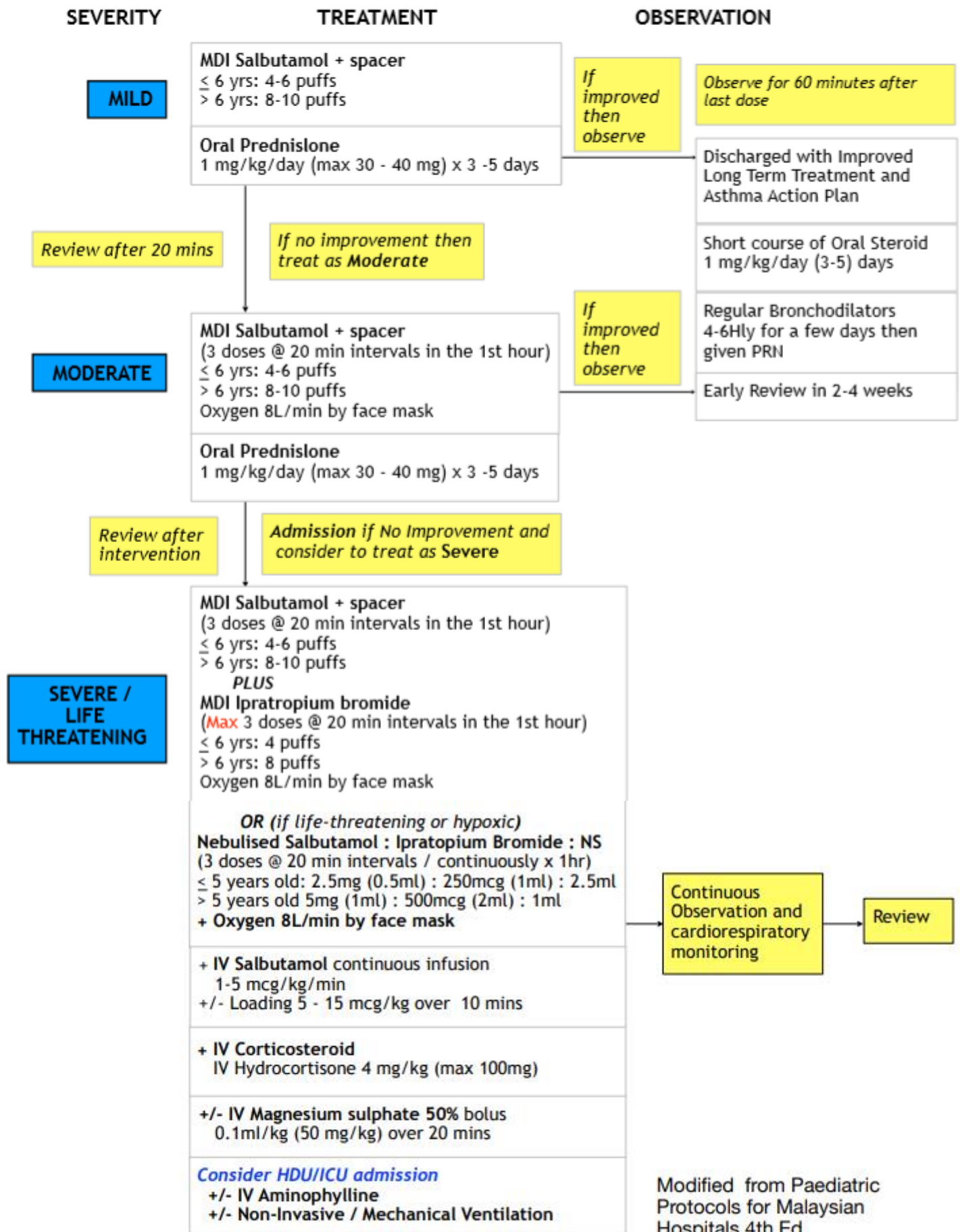
C. Next reassessment within 4-6 hours. Date: _____ Time: _____ PAS: _____ PEFR: _____		
☐ _____ Mild (5 - 7)	☐ Consider trial wean off oxygen ☐ Salbutamol interval can be stretched to 4hourly	
☐ _____ Moderate (8 – 11)	MDI Salbutamol + Spacer (every 4 hour) ☐ ≤6 yo: 4-6 puffs ☐ >6 yo: 8-10 puffs.	
☐ _____ Severe/life threatening (12 – 15)	☐ Arrange for admission	
D. Final reassessment for DISPOSITION. Date: _____ Time: _____ PAS: _____ PEFR: _____		
☐ _____ Mild (5 - 7)	☐ Score 8-11 _____	☐ Score 12-15 _____
DISCHARGE DISPOSITION ☐ Oral prednisolone _____ mg OD for _____ days ☐ Salbutamol 2 -4 puffs every 4 hourly until well ☐ Review asthma action plan ☐ Review MDI and spacer technique ☐ F/up with primary care physician at nearest health clinic for assessment and/or referral to paed clinic in 2-4weeks		ADMISSION ☐ Consider alternative ddx and continue management ☐ Consider admission if no improvement Admission: ☐ Spoken to: _____ at _____ (time) ☐ Admit to: _____

Paediatric Asthma Score (PAS) Parameter		Asthma Score		
		1 point	2 points	3 points
Respiratory rate	2 - 4 yrs	≤ 34	35 - 39	≥ 40
	4 - 5 yrs	≤ 30	31 - 35	≥ 36
	6 - 12 yrs	≤ 26	27 - 30	≥ 31
Oxygen saturation (%)		> 95 with RA	90 - 95 with RA	< 90 with RA or supplemental O2
Breath sounds *If asymmetric lung finding, the most severe is rated		Normal breathing or end exp. wheeze	Expiratory wheeze	Ins. & exp. wheeze, reduced breath sound, or both
Retractions		Non or intercostal	Intercostal & substernal	Intercostal, substernal & supraclavicular
Dyspnea		Speak in sentences	Speaks in partial sentences or short cries	Speaks in single words or short phrases or grunts
Severity of Asthma				
		Mild	Moderate	Severe
Score		5 - 7	8 - 11	12 - 15

ASTHMA DRUGS DILUTION GUIDE

MEDICATION	AGE (YEARS)	
MDI	≤ 6 YEARS	> 6 YEARS
MDI SALBUTAMOL + SPACER 1 PUFF = 100mcg May repeat every 20 min x 3 doses in the 1 st hour	4 – 6 PUFFS	8 – 10 PUFFS
MDI IPRATROPIUM BROMIDE + SPACER (ATROVENT) 1 PUFF = 20mcg May repeat every 20 min x 3 doses in the 1 st hour * only to add-on in severe exacerbation of asthma	4 PUFFS	8 PUFFS
NEBULISER	≤ 5 YEARS	> 5 YEARS
NEB SALBUTAMOL : ATROVENT : NS * only for Life-threatening asthma or hypoxia	SALBUTAMOL 0.5ml (2.5mg) + IPRATROPIUM BROMIDE (1ml) 250mcg + NORMAL SALINE 2.5ml	SALBUTAMOL 1ml (5mg) + IPRATROPIUM BROMIDE 2ml (500mcg) + NORMAL SALINE 1ml
STEROIDS	ALL AGES	
ORAL PREDNISOLONE 1 TAB = 5mg	1 mg/kg	
	MAX	
	< 2 years: 10mg	2-5 years: 20mg
IV HYDROCORTISONE	> 5 years: 30-40mg	
	4 mg/kg/dose (MAX 100mg) 6 hourly	

Management of Acute Exacerbation of Bronchial Asthma in Children



Modified from Paediatric Protocols for Malaysian Hospitals 4th Ed